

Lablet 2 Solution

Instructions

Update configurations on **R1** and **Sw2** by performing the following tasks:

Note: The two desktop hosts are available for testing and verification of work. Server1 can be used as a destination for tests using ping, telnet, and SSH clients from the desktops. Do not change configuration on these hosts.

Task 1: Layer 2 Network Segmentation

- Create VLANs from the table on Sw2.
- Configure network access for the desktop hosts, do not use trunking

Task 2: Inter-VLAN Routing

- Configure an 802.1Q trunk between R1 and Sw2.
- Verify the default VLAN is untagged on the trunk
- Configure R1 to provide inter-VLAN routing between all networks listed in the table

Task 3: Network Address Translation

- Configure R1 to translate traffic from Sw2 networks to the "Internet" using interface PAT.
 - Use pre-configured Access-list 10 to identify traffic to be translated. Do not modify this ACL
- Verify R1's interface IP address is the Inside Global IP address for all translated traffic

Task 4: IPv6 Connectivity

- Configure IPv6 addresses on R1 interfaces using information below
- Configure a static IPv6 default route to the global unicast address on Internet interface E0/1
- Ensure IPv6 connectivity to Server1 using ping from R1.

Device	Interface	IPv6 Address
Internet	Eth0/1	fe80::1/10
Internet	Eth0/1	2001:db8:1::1/64
R1	Eth0/1	2001:db8:1::/64 EUI 64
R1	Eth0/0	fe80::1/10
R1	Eth0/0	2001:db8:10::1/64

Task 1 - Layer 2 Network Segmentation

Sw2

```
enable
config t
!
vlan 13
 name blue
 exit
vlan 35
```

```

name green
exit
vlan 52
name red
exit
vlan 76
name purple
exit
!
interface e0/1
switchport mode access
switchport access vlan 52
no shut
interface e0/2
switchport mode access
switchport access vlan 13
no shut
!
end

```

Verification

Sw2

show vlan brief

VLAN	Name	Status	Ports
1	default	active	Et0/0, Et0/3
13	blue	active	Et0/2
35	green	active	
52	red	active	Et0/1
76	purple	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

show vlan brief

VLAN	Name	Status	Ports
1	default	active	Et0/0, Et0/3
13	blue	active	Et0/2
35	green	active	
52	red	active	Et0/1
76	purple	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Task 2 - Inter-VLAN Routing

Sw2

```

enable
config t

```

```

!
interface e0/0
  switchport trunk encap dot1q
  switchport mode trunk
  no shut
end

```

R1

```

enable
config t
!
interface e0/0
  no shut
interface e0/0.13
  encap dot1q 13
  ip address 10.0.13.1 255.255.255.0
interface e0/0.35
  encap dot1q 35
  ip address 10.0.35.1 255.255.255.0
interface e0/0.52
  encap dot1q 52
  ip address 10.0.52.1 255.255.255.0
interface e0/0.76
  encap dot1q 76
  ip address 10.0.76.1 255.255.255.0
end

```

Verification

Sw2

```
show inter trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/0	on	802.1q	trunking	1
Port Et0/0	Vlans allowed on trunk 1-4094			
Port Et0/0	Vlans allowed and active in management domain 1,13,35,52,76			
Port Et0/0	Vlans in spanning tree forwarding state and not pruned 1,13,35,52,76			

R1

```
show ip int bri
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	unassigned	YES	TFTP	up	up
Ethernet0/0.13	10.0.13.1	YES	manual	up	up
Ethernet0/0.35	10.0.35.1	YES	manual	up	up
Ethernet0/0.52	10.0.52.1	YES	manual	up	up
Ethernet0/0.76	10.0.76.1	YES	manual	up	up
Ethernet0/1	192.0.2.4	YES	TFTP	up	up
Ethernet0/2	unassigned	YES	TFTP	administratively down	down

```
Ethernet0/3          unassigned      YES TFTP    administratively down down
```

```
show interface | inc Ethernet0\0|Encapsulation 802
```

```
Ethernet0/0 is up, line protocol is up  
  Encapsulation 802.1Q Virtual LAN, Vlan ID  1., loopback not set  
Ethernet0/0.13 is up, line protocol is up  
  Encapsulation 802.1Q Virtual LAN, Vlan ID  13.  
Ethernet0/0.35 is up, line protocol is up  
  Encapsulation 802.1Q Virtual LAN, Vlan ID  35.  
Ethernet0/0.52 is up, line protocol is up  
  Encapsulation 802.1Q Virtual LAN, Vlan ID  52.  
Ethernet0/0.76 is up, line protocol is up  
  Encapsulation 802.1Q Virtual LAN, Vlan ID  76.
```

Task 3 - Network Address Translation

R1

```
enable  
config t  
!  
ip nat inside source list 10 interface e0/1 overload  
!  
interface e0/1  
  ip nat outside  
interface e0/0.13  
  ip nat inside  
interface e0/0.35  
  ip nat inside  
interface e0/0.52  
  ip nat inside  
interface e0/0.76  
  ip nat inside  
!  
end
```

Verification

Part 1: R1 Configuration

This is enough to verify that task completed successfully to earn points. Functionality of PAT requires other tasks being completed correctly as well.

```
show run | inc interface Ethernet0\[01]|nat
```

```
interface Ethernet0/0  
interface Ethernet0/0.13  
  ip nat inside  
interface Ethernet0/0.35  
  ip nat inside  
interface Ethernet0/0.52  
  ip nat inside  
interface Ethernet0/0.76  
  ip nat inside  
interface Ethernet0/1  
  ip nat outside  
ip nat inside source list 10 interface Ethernet0/1 overload
```

Note: This verification requires Task 2 having been completed successfully.

```
show ip interface | inc Ethernet|translation
```

```
Ethernet0/0 is up, line protocol is up
Ethernet0/0.13 is up, line protocol is up
  Network address translation is enabled, interface in domain inside
Ethernet0/0.35 is up, line protocol is up
  Network address translation is enabled, interface in domain inside
Ethernet0/0.52 is up, line protocol is up
  Network address translation is enabled, interface in domain inside
Ethernet0/0.76 is up, line protocol is up
  Network address translation is enabled, interface in domain inside
Ethernet0/1 is up, line protocol is up
  Network address translation is enabled, interface in domain outside
Ethernet0/2 is administratively down, line protocol is down
Ethernet0/3 is administratively down, line protocol is down
```

Part 2: Operational Testing

This test will only work if Tasks 1 and 2 are completed successfully. Credit for the points only requires the R1 configuration verification to pass.

Generate traffic from Desktops to populate NAT table

Desk-Red

```
ssh -l cisco 192.0.2.12
```

```
Password: cisco
```

```
Server1>
```

Desk-Blue

```
telnet 192.0.2.12
```

```
Trying 192.0.2.12 ... Open
```

```
User Access Verification
```

```
Username: cisco
```

```
Password: cisco
```

```
Server1>
```

R1

Verify NAT working as expected

```
show ip nat trans
```

Pro	Inside global	Inside local	Outside local	Outside global
tcp	192.0.2.4:4791	10.0.13.101:58824	192.0.2.12:23	192.0.2.12:23
tcp	192.0.2.4:4790	10.0.52.101:11715	192.0.2.12:22	192.0.2.12:22

Disconnect desktops to close sessions

Desk-Red & Desk-Blue

```
exit
```

Task 4 - IPv6 Connectivity

R1

```
enable
config t
!
interface e0/0
  ipv6 address fe80::1 link-local
  ipv6 address 2001:db8:10::1/64
  exit
!
interface e0/1
  ipv6 address 2001:db8:1::/64 eui-64
  exit
!
ipv6 route ::/0 2001:db8:1::1
end
```

Verification

Part 1: Address Configuration

R1

```
show ipv6 interface e0/1 | inc enabled, link-local|2001
```

```
IPv6 is enabled, link-local address is FE80::A8BB:CCFF:FE00:1D10
2001:DB8:1:0:A8BB:CCFF:FE00:1D10, subnet is 2001:DB8:1::/64 [EUI]
```

```
show ipv6 interface e0/0 | inc enabled, link-local|2001
```

```
IPv6 is enabled, link-local address is FE80::1
2001:DB8:10::1, subnet is 2001:DB8:10::/64
```

Part 2: IPv6 Routing / Connectivity

```
show ipv6 route static
```

IPv6 Routing Table - default - 6 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, HG - NHRP registered

Hg - NHRP registration summary, HE - NHRP External, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination

NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter

OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1

ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations

ld - LISP dyn-eid, la - LISP away, le - LISP extranet-policy

lp - LISP publications, ls - LISP destinations-summary, a - Application

m - OMP

```
S ::/0 [1/0]
  via 2001:DB8:1::1
```

```
ping 2001:db8:2::12 repeat 5
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:DB8:2::12, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms